# Aksh Wadhawan

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## Education

**Indian Institute of Information Technology, Jabalpur** Expected May 2028
*B.Tech in Computer Science and Engineering; CGPA: 9.4 / 10.0* *Jabalpur, India*

- CBSE Class 12: 94% | CBSE Class 10: 96.6% | JEE Main 2024: 99.11 percentile

## Technical Skills

**Languages:** C++, Python, JavaScript, C, Dart
**Databases:** MongoDB, MySQL, Redis, ChromaDB (Vector)
**Frameworks:** React.js, Express.js, FastAPI, Flutter, LangGraph, LangChain
**Runtimes & Tools:** Node.js, BullMQ

## Competitive Programming

- Peak Codeforces rating 1773 (Expert); Ranked 199th globally in Codeforces Round 1103 (22,000+ participants).
- Peak LeetCode rating 2128 (Guardian); Ranked 393rd globally in LeetCode Biweekly Contest 187 (39,000+ participants). 1800+ Data Structures and Algorithms (DSA) problems solved across platforms.
- 1st place among 600+ eligible participants in a college-level competitive programming contest.
- 2nd place in CodeRumble 4.0, a coding contest organised by the Programming Club of IIIT Jabalpur.
- National Semi-Finalist, Flipkart GRiD 8.0

## Projects

**Autonomous Patent FTO Analysis System** May 2026 – Present
Live: patent-fto-analyzer.vercel.app | github.com/w-2006-aksh/patent-fto-analyzer

- **Tech Stack:** Python, LangGraph, FastAPI, React, ChromaDB, RAG, EPO OPS API, LLMs
- Built an 11-stage agentic AI pipeline in LangGraph that automates Freedom-to-Operate (FTO) patent analysis: an LLM decomposes each invention into 6 EPO search queries, patents are retrieved from the EPO OPS API and filtered for relevance, an autonomous retry loop generates novel queries on low-yield runs, and surviving patents are embedded and assessed for infringement risk before a legal-style report is produced after human review.
- Parallelized EPO OPS retrieval across a 3-worker thread pool with a custom mutex-locked rate limiter (~2.5 req/s), cutting fetch latency 3x vs. sequential execution within API limits; merged results into a core-first shortlist and built retrieval-augmented generation on ChromaDB using claim-level vector search with per-patent top-k retrieval to balance analysis across patents.
- Implemented element-mapping infringement risk assessment using context-aware LLM prompts for full-claim, partial-claim, and abstract-only patents; output is split into significant-risk findings and cleared patents for auditability.
- Shipped a two-phase product flow with a human-in-the-loop (HITL) checkpoint: the FastAPI backend pauses for user approval of preliminary risk scores before generating the final report, with a React frontend; added sliding-window TPM (tokens-per-minute) budgeting that tracks token usage in a rolling 60s window and throttles LLM calls to stay within free-tier limits.

**Freight Management Platform** July 2025 – Present
Demo: youtu.be/e_ZwZLEQi1I | github.com/w-2006-aksh/freight-management-platform | github.com/w-2006-aksh/freight-driver-app

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, Flutter, Redis, BullMQ
- Built an invite-only freight marketplace where each client curates a trusted-transporter list; implemented backend-enforced bid-invitation access control lists (ACLs) so only authorized transporters can quote on live shipments, with guards against expired bids, duplicate quotes, and cross-client access attempts, across the full shipment lifecycle from posting to delivery.
- Architected separate authentication for the web portals (clients and transporters) and the driver mobile app; winning transporters submit driver details, vehicle number, license, and registration documents via structured upload forms for manual client review before the trip is activated.
- Developed a Flutter mobile driver app launched via JWT-secured deep links from SMS; the app persists active-trip state locally for unreliable connectivity and periodically uploads GPS coordinates to a tracking pipeline that reverse-geocodes locations into a city-level journey timeline on the web dashboard.
- Designed OTP (one-time password) gated delivery confirmation requiring the client to share a verification code with the driver at the destination, preventing premature trip closure.
- Integrated SMS notifications at key shipment lifecycle events via Redis-backed BullMQ job queue workers, decoupling API response time from messaging latency.

## Certifications

- **IBM RAG and Agentic AI Certification**: 9-course series on advanced RAG, vector databases, and agentic workflows.
- **Machine Learning Specialization (DeepLearning.AI & Stanford)**: Supervised, Unsupervised, and Reinforcement Learning (Andrew Ng).